

ering recent disruptions in rare earth supply chains and rising costs, they are being rediscovered owing to their affordability, abundance, and independence from geopolitically sensitive materials. “The global ferrite magnet production is 5–10x that of neodymium. They’re a commodity now,” says Somani.

Compared to neodymium-based magnets, ferrite magnets are cheaper too. Somani says ferrite magnets cost \$5 per kg, while those made of neodymium cost \$60 to \$100. Ferrites are cheaper than even copper, which is around \$11 per kg.

Another equally compelling driver is the pursuit of more sustainable alternatives. “These rare earths are very often found alongside low-grade radioactive elements such as thorium. And therefore, their mining leads to the creation of radioactive tailings,” says Sterling GTake’s Wadhwa.

“If you’re committed to green technology, there’s a clear dichotomy: on the one side, we reduce tailpipe emissions, but on the other, we create environmental hazards through rare earth mining and refining. That contradiction was something we wanted to address,” he adds.

RISE OF INDUCTION MAGNETS

A number of automakers are also exploring technologies that don’t depend on permanent magnets. Jitin Makkar, Senior Vice President and Group Head at rating agency ICRA, says certain OEMs are exploring induction motors in place of permanent magnet motors.

“This is not a reaction to what’s been happening in rare earth metals. This was anyway happening. So, that’s one technology. Globally, material science research keeps happening and companies keep trying to explore multiple materials. They are exploring certain materials that have the same properties,” says Makkar.

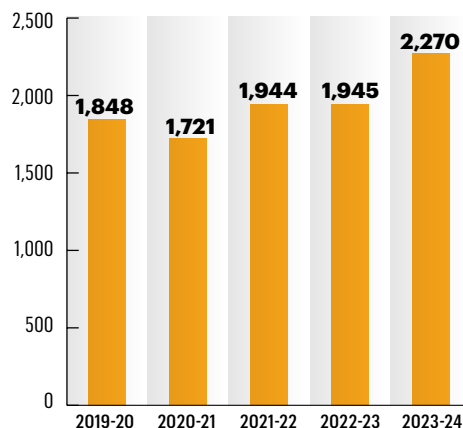


“We have floated the rare earth magnet processing scheme for inter-ministerial consultations”

KAMRAN RIZVI
SECRETARY, MINISTRY
OF HEAVY INDUSTRIES

EV DREAMS

- India’s rare earth imports have been rising over the years as electric vehicles take off
- By 2030, India expects 30% EV penetration



IN METRIC TONNES

SOURCE US GEOLOGICAL SURVEY

However, even if rare earth-free materials prove viable, OEMs will have to go through full validation and testing cycles. “Some may already be working on this as a contingency but their progress isn’t publicly known,” says Makkar.

BECOMING SELF-RELIANT

According to PIB, India’s imports of permanent magnets jumped from 1,945 tonnes in FY23 to 2,270 tonnes in FY24. About 52 India-based companies depend on Chinese magnets for supplying to automobile manufacturers.

China accounted for 69% of total mine production of rare earths in 2024 at 2,70,000 tonnes. It is followed by the US (12%), and Myanmar (8%).

Amidst the increase in dependency on China, the Ministry of Heavy Industries has proposed a rare earth magnet process-